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CBSE CLASS 10 | MATHEMATICS

TRIANGLES

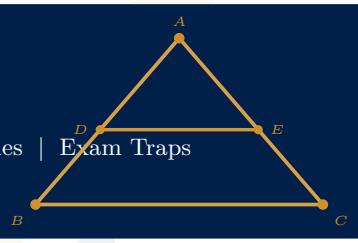
Master Concept, Formula & Visual Problem-Solving Handbook

Academic Year 2026–27

See the ratio. Establish the parallelism.
Prove similarity with precision.

Thales' Theorem (BPT) | AAA, SSS, SAS Criteria | Linear Proportionalities | Exam Traps

A student-friendly Maths Manthan revision resource



What this handbook includes

- Full formal proof of **Basic Proportionality Theorem (BPT)** and its corollaries.
- Rigorous criteria for similarity: **AAA, AA, SSS, and SAS**.
- Linear dimensions: Altitudes, medians, and perimeter invariance (k).
- Classic board models: Trapezium diagonals, shadows, and right-triangle altitude decomposition.

How to use this handbook

Always state the similarity criterion before writing side ratios. Write triangle vertices in **strict corresponding order** ($\triangle ABC \sim \triangle DEF$). State the formal justification (e.g. *"By BPT"* or *"By AA similarity"*) at every step to secure full step marks.

Official CBSE Curriculum Scope (Academic Year 2026–27)

Strict Syllabus Integrity & Deleted Content: Under the rationalized CBSE 2026–27 curriculum, the **Area of Similar Triangles Theorem** ($\frac{\text{Area}_1}{\text{Area}_2} = (\frac{s_1}{s_2})^2$) and the **Formal Proof of Pythagoras Theorem** (and its converse) have been **permanently excised** from Chapter 6. This handbook covers the complete 100% prescribed syllabus: BPT, Converse of BPT, and the Similarity Criteria (AAA, SSS, SAS). Source: CBSE Secondary Curriculum (2026–27), Unit IV: Geometry.

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Prepared for learning and revision – not an official CBSE publication.

CHAPTER 06

SUPPORTING SKILL

GEOMETRIC FOUNDATIONS: SIMILARITY

Congruence vs. Similarity, The Scale Factor & Vertex Correspondence Notation

Module 1.1 | The Conceptual Leap: Congruence vs. Similarity

Two geometric figures are:

- **Congruent (≅):** If they have the **same shape** and the **same size**. Superimposing one on the other results in a perfect match. (Scale factor $k = 1$).
- **Similar (∼):** If they have the **same shape**, but not necessarily the same size. One is an exact enlargement or reduction of the other. (Scale factor $k > 0$).
- **Core Axiom:** *All congruent figures are similar, but similar figures need not be congruent.*

Module 1.2 | General Conditions for Similarity of Polygons

Two polygons with the same number of sides are **similar** if and only if:

(i) All corresponding angles are **equal** (Equiangular).

(ii) All corresponding sides are in the **same ratio** (Proportional).



Module 1.3 | The Golden Rule of Similarity Notation: Vertex Ordering

In board examinations, writing the vertices in correct correspondence order is mandatory:

If $\triangle$**ABC** $\sim$ $\triangle$**DEF**, it means:

$$\boxed{\angle A = \angle D, \quad \angle B = \angle E, \quad \angle C = \angle F} \quad \text{and} \quad \boxed{\frac{AB}{DE} = \frac{BC}{EF} = \frac{CA}{FD} = k}$$

where k is the **scale factor** (or similarity ratio).

Critical Board Presentation Trap: Misaligned Letters

Never write $\triangle ABC \sim \triangle DFE$ if $\angle B = \angle E$ and $\angle C = \angle F$. Swapping vertex letters changes the implied corresponding sides and results in automatic deduction of marks in CBSE step-evaluation.

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CHAPTER 06

CORE SYLLABUS

BASIC PROPORTIONALITY THEOREM (BPT)

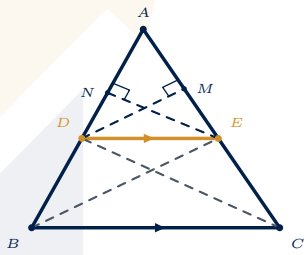
Thales' Theorem: Formal Statement, Construction & Step-by-Step Board Proof

Module 2.1 | Official CBSE Theorem Statement

CBSE Theorem 6.1 (Basic Proportionality Theorem / Thales' Theorem):
If a line is drawn parallel to one side of a triangle to intersect the other two sides in distinct points, the other two sides are divided in the same ratio.

$$\text{In } \triangle ABC,\ DE \parallel BC \implies \frac{AD}{DB} = \frac{AE}{EC}$$

Module 2.2 | Standard CBSE Board Proof (The Area Ratio Method)



Given: $\triangle ABC$ with line $DE \parallel BC$ intersecting AB at D, AC at E.
To Prove: $\frac{AD}{DB} = \frac{AE}{EC}$.
Construction: Join BE and CD. Draw $EN \perp AB$ and $DM \perp AC$.

Proof Steps:

1. $\text{Area}(\triangle ADE) = \frac{1}{2} \cdot AD \cdot EN$, $\quad \text{Area}(\triangle BDE) = \frac{1}{2} \cdot DB \cdot EN$.
   (EN is the altitude to base AB for both triangles).
2. Ratio: $\frac{\text{Area}(\triangle ADE)}{\text{Area}(\triangle BDE)} = \frac{\frac{1}{2} \cdot AD \cdot EN}{\frac{1}{2} \cdot DB \cdot EN} = \frac{AD}{DB}$ $\quad$ — (1)
3. Similarly on base AC: $\frac{\text{Area}(\triangle ADE)}{\text{Area}(\triangle CDE)} = \frac{\frac{1}{2} \cdot AE \cdot DM}{\frac{1}{2} \cdot EC \cdot DM} = \frac{AE}{EC}$ $\quad$ — (2)
4. $\triangle BDE$ and $\triangle CDE$ lie on the **same base** DE and between the **same parallels** $DE \parallel BC$:

$$\text{Area}(\triangle BDE) = \text{Area}(\triangle CDE) \quad \text{— (3)}$$

5. From (1), (2), and (3): $\mathbf{\frac{AD}{DB} = \frac{AE}{EC}}$. **[Hence Proved]**

Pedagogical Keystone: Why Do Oblique Altitudes Work?

Notice that $\triangle BDE$ is obtuse-angled. Its altitude from vertex E falls on the line extension of base BD (segment AB). Thus EN serves simultaneously as the perpendicular altitude for both acute $\triangle ADE$ and obtuse $\triangle BDE$.

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CORE SYLLABUS

BPT COROLLARIES & CONVERSE OF THALES

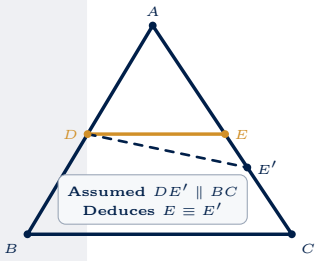
The Three Algebraic Formulations, Proof by Contradiction & Mid-Point Specialization

Module 3.1 | The Three Master Algebraic Forms of BPT

By applying basic ratio operations (componendo), two additional forms emerge directly:

Form	Algebraic Identity	Application in Board Exam Problems
Form 1 (Standard)	$\frac{AD}{DB} = \frac{AE}{EC}$	Direct ratio of upper segments to lower segments. Used when all 4 segments are given.
Form 2 (Part-to-Whole)	$\frac{AD}{AB} = \frac{AE}{AC}$	Add 1: $\frac{AD}{DB} + 1 = \frac{AE}{EC} + 1 \implies \frac{AB}{DB} = \frac{AC}{EC}$. Inverting gives $\frac{AD}{AB} = \frac{AE}{AC}$.
Form 3 (Lower-to-Whole)	$\frac{DB}{AB} = \frac{EC}{AC}$	Compares bottom segments directly to the entire side lengths AB and AC.

Module 3.2 | Converse of Basic Proportionality Theorem (Theorem 6.2)



CBSE Theorem 6.2 (Converse of BPT):
If a line divides two sides of a triangle in the same ratio, then the line is parallel to the third side.

$$\text{In } \triangle ABC, \quad \frac{AD}{DB} = \frac{AE}{EC} \implies DE \parallel BC$$

Proof by Contradiction Summary:

1. Suppose line DE is **not** parallel to BC.
2. Draw line $DE' \parallel BC$ meeting AC at E'.
3. By BPT: $\frac{AD}{DB} = \frac{AE'}{E'C}$. But given: $\frac{AD}{DB} = \frac{AE}{EC}$.
4. Hence: $\frac{AE}{EC} = \frac{AE'}{E'C} \implies \frac{AE+EC}{EC} = \frac{AE'+E'C}{E'C} \implies \frac{AC}{EC} = \frac{AC}{E'C}$.
5. Thus $EC = E'C$, which means E **and** E' **coincide**.
6. Therefore, $DE \parallel BC$. **[Q.E.D.]**

Module 3.3 | Mid-Point Theorem as a Direct Specialization of BPT

Connecting Class 9 and Class 10 Geometry

If D and E are mid-points of AB and AC, then $AD = DB$ and $AE = EC$, so $\frac{AD}{DB} = \frac{AE}{EC} = 1$. By the **Converse of BPT**, $DE \parallel BC$. Thus, the Class 9 Mid-Point Theorem is simply BPT at $k = 1$.

CHAPTER 06

WORKED MODEL

BPT PROBLEM-SOLVING & TRAPEZIUM MODELS

Algebraic Protocols for Unknown x & The Diagonal Proportionality Theorem

Module 4.1 | The Algebraic Protocol for Solving BPT Problems

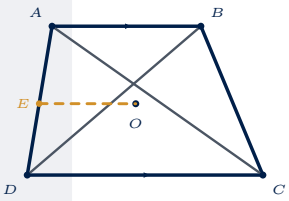
- **Step 1:** Verify that $DE \parallel BC$ is stated. **Step 2:** Write $\frac{AD}{DB} = \frac{AE}{EC}$ citing *"(By BPT)"*.
- **Step 3:** Substitute algebraic expressions, cross-multiply, and solve the equation.
- **Step 4:** Reject extraneous negative roots (segment lengths must remain strictly positive).

Module 4.2 | Worked Archetype 1: Solving Quadratic BPT Equations

Worked Archetype 1: Determining Variable x Using BPT

Problem: In $\triangle ABC$, $DE \parallel BC$. If $AD = x$, $DB = x - 2$, $AE = x + 2$, and $EC = x - 1$, find x.
Solution: By **Basic Proportionality Theorem**: $\frac{AD}{DB} = \frac{AE}{EC} \implies \frac{x}{x-2} = \frac{x+2}{x-1}$.
Cross-multiplying: $x(x-1) = (x-2)(x+2) \implies x^2 - x = x^2 - 4 \implies -x = -4 \implies \mathbf{x = 4}$ cm.

Module 4.3 | The Classic Trapezium Proportionality Theorem



CBSE Standard 3-Mark Theorem:
The diagonals of a trapezium divide each other proportionally.

In trapezium $ABCD$ with $AB \parallel CD$, $\quad \dfrac{\mathbf{AO}}{\mathbf{OC}} = \dfrac{\mathbf{BO}}{\mathbf{OD}}$

Proof Construction: Through point O, draw line $OE \parallel AB \parallel CD$ meeting AD at E.

1. In $\triangle ADC$, since $OE \parallel CD$: $\frac{AE}{ED} = \frac{AO}{OC}$ — (1) [By BPT]
2. In $\triangle DAB$, since $EO \parallel AB$: $\frac{ED}{AE} = \frac{OD}{BO} \implies \frac{AE}{ED} = \frac{BO}{OD}$ — (2)
3. Equating (1) and (2): $\mathbf{\frac{AO}{OC} = \frac{BO}{OD}}$. **[Q.E.D.]**

Converse: If $\frac{AO}{OC} = \frac{BO}{OD}$, then $ABCD$ is a **trapezium** ($AB \parallel CD$).

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CHAPTER 06

CORE SYLLABUS

SIMILARITY CRITERIA: AAA & AA

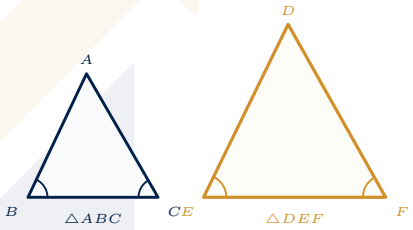
Equiangularity Mechanics, The 2-Angle Board Shortcut & Shadow Models

Module 5.1 | The AAA Similarity Criterion (Theorem 6.3)

CBSE Theorem 6.3 (AAA Similarity Criterion):
If in two triangles, corresponding angles are equal, then their corresponding sides are in the same ratio, and hence the two triangles are similar.

$$\boxed{\angle A = \angle D, \quad \angle B = \angle E, \quad \angle C = \angle F \implies \triangle ABC \sim \triangle DEF}$$

Module 5.2 | The Indispensable AA Similarity Shortcut (Board Exam Standard)



In any triangle, the sum of all three interior angles is $180°$.
If two angles of $\triangle ABC$ equal two angles of $\triangle DEF$:

$$\angle A = \angle D \quad \text{and} \quad \angle B = \angle E$$

Then the third angles are automatically equal:

$$\angle C = 180° - (\angle A + \angle B) = 180° - (\angle D + \angle E) = \angle F$$

Golden Exam Rule: Use AA Similarity

Proving **any two pairs of corresponding angles equal** is sufficient to declare similarity by the **AA criterion**.

Module 5.3 | Worked Archetype 2: The Vertical Pole & Tower Shadow Problem

Worked Archetype 2: Tower and Pole Shadow Model

Problem: A vertical pole of 6 m casts a shadow of 4 m. At the same time, a tower casts a shadow of 28 m. Find the tower's height h.
Solution: $\angle B = \angle Q = 90°$ and $\angle C = \angle R$ (Sun's elevation is identical at same time). By **AA similarity**, $\triangle ABC \sim \triangle PQR \implies \frac{AB}{PQ} = \frac{BC}{QR} \implies \frac{6}{h} = \frac{4}{28} \implies \frac{6}{h} = \frac{1}{7} \implies \mathbf{h = 42}$ m.

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CHAPTER 06

CORE SYLLABUS

SIMILARITY CRITERIA: SSS & SAS

Side-Ratio Equiangularity, The Included-Angle Rule & The SSA Trap Alert

Module 6.1 | The SSS Similarity Criterion (Theorem 6.4)

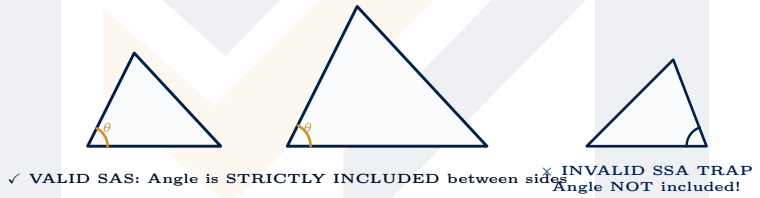
CBSE Theorem 6.4 (SSS Similarity Criterion):
If in two triangles, sides of one triangle are proportional to the sides of the other triangle, then their corresponding angles are equal and hence the two triangles are similar.

$$\boxed{\frac{AB}{DE} = \frac{BC}{EF} = \frac{CA}{FD} = k \implies \triangle ABC \sim \triangle DEF}$$

Module 6.2 | The SAS Similarity Criterion (Theorem 6.5) & The "Included Angle" Rule

CBSE Theorem 6.5 (SAS Similarity Criterion):
If one angle of a triangle equals one angle of another triangle and the sides including these angles are proportional, then the two triangles are similar.

$$\boxed{\angle A = \angle D \quad \text{and} \quad \frac{AB}{DE} = \frac{AC}{DF} \implies \triangle ABC \sim \triangle DEF}$$



Module 6.3 | Worked Archetype 3: The Intersecting Chords / Butterfly Configuration

Worked Archetype 3: Intersecting Diagonals Model

Problem: Line segments AB and CD intersect at O with $OA = 4$ cm, $OB = 6$ cm, $OC = 8$ cm, and $OD = 12$ cm. Prove $\triangle AOC \sim \triangle BOD$.
Solution: Compare side ratios: $\frac{OA}{OB} = \frac{4}{6} = \frac{2}{3}$ and $\frac{OC}{OD} = \frac{8}{12} = \frac{2}{3} \implies \frac{OA}{OB} = \frac{OC}{OD}$.
Included angles: $\angle AOC = \angle BOD$ (vertically opposite). By **SAS similarity**: $\triangle$**AOC** $\sim \triangle$**BOD**.

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CHAPTER 06

WORKED MODEL

LINEAR PROPORTIONALITIES IN SIMILAR TRIANGLES

Altitudes, Medians, Perimeters & The Master Linear Invariance Ratio

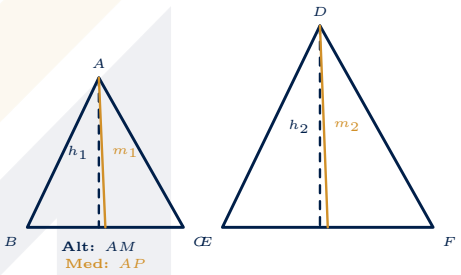
Module 8.1 | The Master Linear Invariance Theorem

When two triangles are similar, the scale factor k governs **every linear measure** of the triangles:

If $\triangle$**ABC** $\sim \triangle$**DEF** with scale factor $k = \frac{AB}{DE} = \frac{BC}{EF} = \frac{CA}{FD}$:

$$\boxed{\frac{\text{Perimeter}(\triangle ABC)}{\text{Perimeter}(\triangle DEF)} = \frac{\text{Altitude } h_1}{\text{Altitude } h_2} = \frac{\text{Median } m_1}{\text{Median } m_2} = \frac{\text{Inradius } r_1}{\text{Inradius } r_2} = \mathbf{k}}$$

Module 8.2 | Standard CBSE Board Proofs (High-Yield 3-Markers)



Theorem A: Ratio of Altitudes = Ratio of Sides
Let $AM \perp BC$ and $DN \perp EF$.

1. In $\triangle ABM$ and $\triangle DEN$: $\angle B = \angle E$ and $\angle AMB = \angle DNE = 90°$.
2. By **AA similarity**: $\triangle ABM \sim \triangle DEN$.
3. Hence: $\mathbf{\frac{AM}{DN} = \frac{AB}{DE} = k}$. **[Q.E.D.]**

Theorem B: Ratio of Medians = Ratio of Sides
Let AP and DQ be medians to BC and EF.

1. Since P, Q are mid-points: $\frac{BP}{EQ} = \frac{\frac{1}{2}BC}{\frac{1}{2}EF} = \frac{BC}{EF} = \frac{AB}{DE}$.
2. Also $\angle B = \angle E$.
3. By **SAS similarity**: $\triangle ABP \sim \triangle DEQ$.
4. Hence: $\mathbf{\frac{AP}{DQ} = \frac{AB}{DE} = k}$. **[Q.E.D.]**

Module 8.3 | Ratio of Perimeters Proof

By Componendo / Ratio Addition: If $\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2} = k$, then:

$$\frac{\text{Perimeter}_1}{\text{Perimeter}_2} = \frac{a_1 + b_1 + c_1}{a_2 + b_2 + c_2} = \frac{ka_2 + kb_2 + kc_2}{a_2 + b_2 + c_2} = \frac{k(a_2 + b_2 + c_2)}{a_2 + b_2 + c_2} = \mathbf{k}.$$

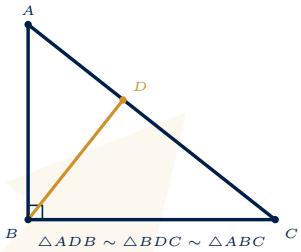
CHAPTER 06

WORKED MODEL

WORKED BOARD-EXAM MASTER MODELS

Right-Triangle Altitude Decomposition & The Walking Person Shadow Model

Module 9.1 | Master Model 1: Right-Triangle Altitude Decomposition



The Geometric Mean Theorem:
In $\triangle ABC$ right-angled at B, with altitude $BD \perp AC$:
1. $\triangle ADB \sim \triangle ABC$ *(By AA: $\angle A$ common, $\angle ADB = 90°$).*
2. $\triangle BDC \sim \triangle ABC$ *(By AA: $\angle C$ common, $\angle BDC = 90°$).*
3. $\triangle$**ADB** $\sim \triangle$**BDC** *(Both similar to $\triangle ABC$).*

Four Master Metric Relations:

$$\boxed{\mathbf{BD^2 = AD \cdot DC}} \quad \text{(Altitude Theorem)}$$

$$\boxed{\mathbf{AB^2 = AD \cdot AC}} \qquad \boxed{\mathbf{BC^2 = CD \cdot AC}}$$

$$\boxed{\mathbf{AB \cdot BC = AC \cdot BD}} \quad \text{(Area Identity)}$$

Module 9.2 | Master Model 2: The Walking Person Shadow Model (CBSE Standard 4-Marker)

Worked Archetype 4: Moving Person Shadow Protocol

Problem: A girl of height 90 cm (0.9 m) walks away from a 3.6 m lamppost at 1.2 m/s. Find shadow length after 4 seconds.
Solution:
1. Distance walked $BD = 1.2$ m/s $\times$ 4 s $= 4.8$ m. Let shadow $DE = s$.
2. In $\triangle ABE$ and $\triangle CDE$: $\angle B = \angle D = 90°$ and $\angle E$ is common.
3. By **AA similarity**: $\triangle ABE \sim \triangle CDE \implies \frac{AB}{CD} = \frac{BE}{DE} \implies \frac{3.6}{0.9} = \frac{4.8+s}{s}$.
4. $4 = \frac{4.8+s}{s} \implies 4s = 4.8 + s \implies 3s = 4.8 \implies \mathbf{s = 1.6}$ m (160 cm).

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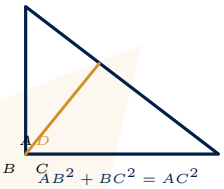
EXTENSION PRACTICE

ENRICHMENT & GEOMETRIC BRIDGES

Pythagoras via Similarity, Internal Angle Bisector & Competitive Edges

Module 10.1 | The 3-Line Similarity Derivation of Pythagoras Theorem

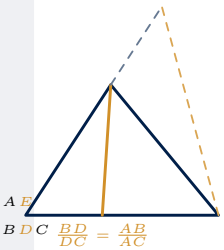
Although the axiomatic textbook proof is deleted from CBSE 2026–27, understanding its similarity derivation bridges geometry with algebra:



The 3-Line Derivation:

1. In right $\triangle ABC$ ($\angle B = 90°$) with altitude $BD \perp AC$:
2. $\triangle ADB \sim \triangle ABC \implies \frac{AB}{AC} = \frac{AD}{AB} \implies \mathbf{AB^2 = AD \cdot AC}$ — (1)
3. $\triangle BDC \sim \triangle ABC \implies \frac{BC}{AC} = \frac{CD}{BC} \implies \mathbf{BC^2 = CD \cdot AC}$ — (2)
4. Adding (1) + (2): $AB^2 + BC^2 = AC(AD + CD) = AC \cdot AC = \mathbf{AC^2}$.

Module 10.2 | The Internal Angle Bisector Theorem (Competitive Bridge)



Internal Angle Bisector Theorem:
The internal bisector of an angle of a triangle divides the opposite side in the ratio of the containing sides.

$$\boxed{\text{In } \triangle ABC,\ AD \text{ bisects } \angle A \implies \frac{\mathbf{BD}}{\mathbf{DC}} = \frac{\mathbf{AB}}{\mathbf{AC}}}$$

Proof via BPT: Draw $CE \parallel DA$ meeting BA produced at E. $\angle AEC = \angle BAD$ (corr.) and $\angle ACE = \angle CAD$ (alt.). Thus $AE = AC$. By BPT on $\triangle BCE$: $\frac{BD}{DC} = \frac{BA}{AE} = \frac{AB}{AC}$.

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REFERENCE MATRIX

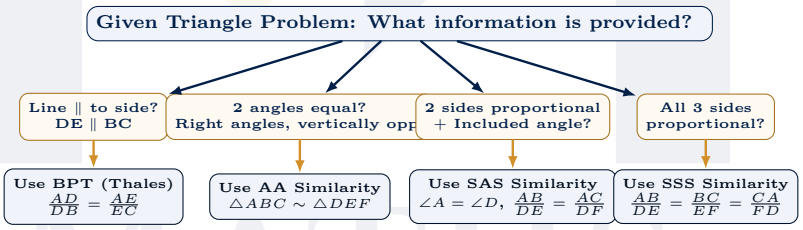
MASTER SUMMARY & DECISION ALGORITHM

Comprehensive Similarity Comparison Matrix & 10-Second Problem Solver

Module 11.1 | Comprehensive Theorem & Criteria Repository

Theorem / Criterion	Core Mathematical Condition	Key Exam Application
BPT (Thales')	$DE \parallel BC \implies \frac{AD}{DB} = \frac{AE}{EC}$	Finding unknown side segments when parallel transversal is given.
Converse of BPT	$\frac{AD}{DB} = \frac{AE}{EC} \implies DE \parallel BC$	Proving a line segment is parallel to a triangle base.
AA Criterion	$\angle A = \angle D,\ \angle B = \angle E$	The most frequent board shortcut (e.g., shadows, vertical lines).
SSS Criterion	$\frac{AB}{DE} = \frac{BC}{EF} = \frac{CA}{FD}$	Proving triangles similar when all 3 side lengths/ratios are given.
SAS Criterion	$\angle A = \angle D,\ \frac{AB}{DE} = \frac{AC}{DF}$	Used when 2 pairs of sides and their strictly **included** angle match.
Trapezium Theorem	$\frac{AO}{OC} = \frac{BO}{OD}$	Calculating intersection segments in trapezium diagonals.
Linear Invariance	$\frac{P_1}{P_2} = \frac{h_1}{h_2} = \frac{m_1}{m_2} = \frac{\text{side}_1}{\text{side}_2}$	Rapid calculation of perimeter, altitude, or median ratios (k).
Right △ Altitude	$BD^2 = AD \cdot DC$	Computing heights/projections in altitude-to-hypotenuse triangles.

Module 11.2 | The 10-Second Similarity Decision Algorithm



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CHAPTER 06

TRAP ALERT

EXAM TRAPS, HACKS & PRE-SUBMISSION AUDIT

Crucial Board Pitfalls, Instant Verification Checks & Final Protocol

Module 12.1 | Top 8 CBSE Board Exam Traps in Triangles

1. **Vertex Order Mismatch in Similarity Statements:** Writing $\triangle ABC \sim \triangle DEF$ implies $\frac{AB}{DE} = \frac{BC}{EF} = \frac{CA}{FD}$. If you write $\triangle ABC \sim \triangle DFE$, your cross-multiplication will be mathematically invalid.
2. **The Non-Included Angle Trap in SAS:** SAS requires the angle to be **between** the proportional sides. If $\frac{AB}{DE} = \frac{AC}{DF}$ and $\angle B = \angle E$, the triangles are NOT necessarily similar!
3. **Applying BPT on Non-Parallel Lines:** You cannot apply $\frac{AD}{DB} = \frac{AE}{EC}$ unless $DE \parallel BC$ is explicitly given or proved first.
4. **Confusing Segment Ratio with Whole Side Ratio in BPT:** BPT states $\frac{AD}{DB} = \frac{AE}{EC}$. Do not substitute AB in place of DB unless using the derived componendo form $\frac{AD}{AB} = \frac{AE}{AC}$.
5. **Inverting Ratios in Trapezium Diagonals:** In trapezium $ABCD$, $\frac{AO}{OC} = \frac{BO}{OD}$. Students often mistakenly write $\frac{AO}{OC} = \frac{OD}{BO}$, reversing the transversal orientation.
6. **Missing Theorem Statements in Proofs:** For BPT in subjective questions, you MUST write the formal *"Given", "To Prove", "Construction"*, and state *"triangles on same base between same parallels have equal areas"*.
7. **Quoting Deleted Theorems:** Do not attempt to solve questions using the deleted Area Ratio Theorem ($\frac{\text{Area}_1}{\text{Area}_2} = k^2$). Stick to perimeter and linear side relations.
8. **Unit Inconsistency in Shadow Questions:** Ensure shadow lengths and pole heights are in the same unit (m or cm) before writing ratios.

Module 12.2 | Arithmetic & Geometric Verification Hacks

- **Cross-Multiplication Symmetry Check:** In $\frac{AD}{DB} = \frac{AE}{EC}$, check that $AD \cdot EC = DB \cdot AE$.
- **Scale Factor Consistency (k):** Compute $\frac{a_1}{a_2}, \frac{b_1}{b_2}, \frac{c_1}{c_2}$ separately; all three decimal quotients must be identical.
- **Positive Segment Filter:** For quadratic equations, reject any root that yields a negative length ($x - 2 < 0$).

Module 12.3 | The 5-Point Pre-Submission Board Exam Audit

Check	Verification Item	Board Evaluation Standard
☐ 1	Vertex Correspondence Checked	Are the triangle vertices ordered so that corresponding angles match?
☐ 2	Similarity Criterion Explicitly Stated	Did you write "(By AA similarity)" or "(By BPT)" next to every step?
☐ 3	SAS Angle is Strictly Included	Is the equal angle located between the two proportional sides?
☐ 4	Parallel Line Justification	Is the parallelism $DE \parallel BC$ clearly justified before invoking BPT?
☐ 5	Final Units Explicitly Stated	Are units (cm, m) explicitly written after all length and shadow answers?

Master the Theorems. Construct the Logic. Secure the 100%.

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